MATH 241: Calculus III

Fall 2026

Lecture 8: Partial Derivatives

Background & Motivation

Now we differentiate. The idea is simple: hold all variables constant except one, and differentiate with respect to that one. This gives us partial derivatives—the multivariate analog of $f'(x)$. Today we define them, compute them, and see how they measure the slope of a surface in each coordinate direction.

1. Partial Derivatives

Definition 1: Partial Derivative (Limit Definition)

The **partial derivative** $f_x(a, b)$ is the instantaneous rate of change of $f(x, y)$ in the x-direction at the point (a, b), with y held fixed at b. Geometrically, slice the graph of f with the vertical plane $y = b$; the resulting cross-section is a single-variable curve $z = f(x, b)$, and $f_x(a, b)$ is the **slope of the tangent line** to that curve at $x = a$. In other words, partial derivatives are ordinary single-variable slopes taken along axis-aligned slices of the surface.

$$f_x(x, y) = \frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

With respect to y:

$$f_y(x, y) = \frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

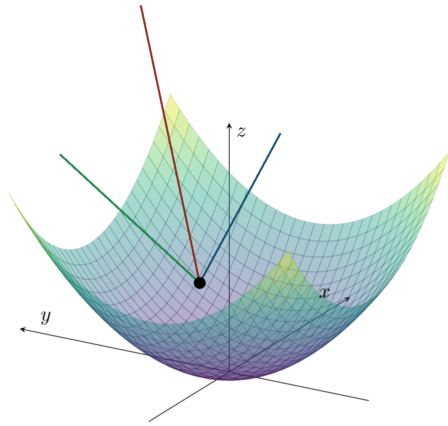
Practical rule: To find f_x, treat y as a constant and differentiate with respect to x. For f_y, treat x as constant.

Geometric Interpretation

– $f_x(a, b)$ is the slope of the tangent line to $z = f(x, b)$ at $x = a$

– Measures rate of change of f in the x-direction while y is held fixed

– Similarly, $f_y(a, b)$ is the slope of $z = f(a, y)$ at $y = b$



Tangent line (red) at $(1, 1, 2)$ on $z = x^2 + y^2$. Blue dashed: xz-projection (slope $= f_x = 2$). Green dashed: yz-projection (slope $= f_y = 2$).

Example 1: Using the Limit Definition

Use the limit definition to find f_x for $f(x, y) = x^2y + 3xy^2$.

1. Set up: $f_x = \lim_{h\to 0} \frac{(x+h)^2y+3(x+h)y^2-x^2y-3xy^2}{h}$

2. Expand: $= \lim_{h\to 0} \frac{x^2y+2xhy+h^2y+3xy^2+3hy^2-x^2y-3xy^2}{h}$

$= \lim_{h\to 0} \frac{2xhy+h^2y+3hy^2}{h}$

3. Simplify and evaluate: $= \lim_{h\to 0}(2xy + hy + 3y^2) = \boxed{2xy + 3y^2}$

Example 2: Computing Partial Derivatives

Find all first partial derivatives:

(a) $f(x, y) = x^3e^y + \sin(xy)$

$f_x = 3x^2e^y + y\cos(xy)$

$f_y = x^3e^y + x\cos(xy)$

(b) $g(x, y) = \dfrac{x^2}{x + y}$

$g_x = \frac{2x(x+y)-x^2}{(x+y)^2} = \frac{x^2+2xy}{(x+y)^2}$ (quotient rule in x)

$g_y = \frac{0-x^2}{(x+y)^2} = \frac{-x^2}{(x+y)^2}$

(c) $h(x, y, z) = xye^z + z\ln(x + y)$

$h_x = ye^z + \frac{z}{x+y}$, $h_y = xe^z + \frac{z}{x+y}$, $h_z = xye^z + \ln(x + y)$

2. Derivative Rules for Partials

Rules (Hold Other Variables Constant)

- **Linearity:** $\partial_x[af + bg] = af_x + bg_x$
- **Product:** $\partial_x[fg] = f_x g + f g_x$
- **Quotient:** $\partial_x[f/g] = (f_x g - f g_x)/g^2$
- **Chain:** $\partial_x[f(g(x,y))] = f'(g) \cdot g_x$
- **Zero partial:** if $f_x = 0$ everywhere $\implies$ f does not depend on x
- **Constants:** $\partial_x[c] = 0$; treat other variables as constants

3. Higher-Order Partial Derivatives

Definition 2: Second-Order Partial Derivatives

A **second-order partial derivative** is a partial derivative of a partial derivative. The two *pure* second partials f_{xx} and f_{yy} measure **concavity along coordinate directions** — how fast the slope f_x changes as you move further in x, and similarly for y — exactly analogous to the second derivative of a single-variable function. The two *mixed* partials f_{xy} and f_{yx} measure something new: how the slope in *one* direction changes as you move in the *other* direction, capturing the graph's twist or saddle-like behavior at a point.

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) \quad \text{(pure, in } x\text{)}$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} \quad \text{(pure, in } y\text{)}$$

$$f_{xy} = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \quad \text{(mixed: } x \text{ first, then } y\text{)}$$

$$f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y} \quad \text{(mixed: } y \text{ first, then } x\text{)}$$

Theorem 1: Clairaut's Theorem (Equality of Mixed Partials)

If f_{xy} and f_{yx} are both continuous on an open region, then:

$$f_{xy} = f_{yx}$$

The order of differentiation does not matter (under mild continuity conditions).

Why Clairaut's Theorem Works (Proof Outline)

– Consider the second-order difference quotient over a small rectangle $[x, x+h] \times [y, y+k]$

– Define $\Delta = f(x+h, y+k) - f(x+h, y) - f(x, y+k) + f(x, y)$

– Apply the Mean Value Theorem twice: first in x, then in y $\implies$ $\Delta = f_{xy}(c_1, d_1) \cdot hk$

– Apply MVT in the other order: first in y, then in x $\implies$ $\Delta = f_{yx}(c_2, d_2) \cdot hk$

– As $h, k \to 0$, continuity gives $f_{xy}(x, y) = f_{yx}(x, y)$

When Clairaut Fails – A Counterexample

– Consider $f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

– One can show $f_{xy}(0,0) = -1$ and $f_{yx}(0,0) = 1$ – the mixed partials are *not equal* at the origin because they are not continuous there

Example 3: Clairaut Verification

Let $f(x,y) = x^3y^2 + e^{xy}$. Find all four second-order partials and verify $f_{xy} = f_{yx}$.

1. First partials:

$f_x = 3x^2y^2 + ye^{xy}, \quad f_y = 2x^3y + xe^{xy}$

2. Second partials:

$f_{xx} = 6xy^2 + y^2e^{xy}$

$f_{yy} = 2x^3 + x^2e^{xy}$

$f_{xy} = \frac{\partial}{\partial y}[3x^2y^2 + ye^{xy}] = 6x^2y + e^{xy} + xye^{xy}$

$f_{yx} = \frac{\partial}{\partial x}[2x^3y + xe^{xy}] = 6x^2y + e^{xy} + xye^{xy}$

3. Result: $f_{xy} = f_{yx}$ ✓

Example 4: Application: Container Design

A cylindrical container has volume $V(r,h) = \pi r^2 h$ and surface area $S(r,h) = 2\pi rh + 2\pi r^2$. If $r = 5$ cm and $h = 10$ cm, how does V change if we increase r by 0.1 cm? If we increase h by 0.1 cm?

1. Compute partials of V:

$V_r = 2\pi rh = 2\pi(5)(10) = 100\pi \approx 314$ cm^3/cm

$V_h = \pi r^2 = 25\pi \approx 78.5$ cm^3/cm

2. Estimate changes:

$\Delta r = 0.1$: $\Delta V \approx V_r \cdot 0.1 = 10\pi \approx 31.4$ cm^3

$\Delta h = 0.1$: $\Delta V \approx V_h \cdot 0.1 = 2.5\pi \approx 7.85$ cm^3

3. Result: Increasing radius has $\sim 4\times$ more effect on volume than increasing height.

Practice Problems

Problem 1. Find f_x and f_y for $f(x,y) = x^2 \cos y + y \ln x$.

Answer: $f_x = 2x \cos y + y/x, \quad f_y = -x^2 \sin y + \ln x$

Problem 2. Find all first partials of $f(x,y,z) = xyz + z^2 \sin(x)$.

$f_x = yz + z^2 \cos x, \quad f_y = xz, \quad f_z = xy + 2z \sin x$

Problem 3. For $f(x,y) = \arctan(y/x)$, show that $f_{xx} + f_{yy} = 0$ (i.e., f is **harmonic**).

$f_x = \frac{-y/x^2}{1+y^2/x^2} = \frac{-y}{x^2+y^2}, \quad f_y = \frac{1/x}{1+y^2/x^2} = \frac{x}{x^2+y^2}$
$f_{xx} = \frac{2xy}{(x^2+y^2)^2}, \quad f_{yy} = \frac{-2xy}{(x^2+y^2)^2}$
$f_{xx} + f_{yy} = 0$ ✓

Problem 4. Use the limit definition to compute $f_x(0,0)$ for $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = \boxed{0}$

Problem 5. Find f_{xxy} for $f(x,y) = e^x \sin(xy)$.
$f_x = e^x \sin(xy) + ye^x \cos(xy)$
$f_{xx} = e^x \sin(xy) + ye^x \cos(xy) + ye^x \cos(xy) - y^2 e^x \sin(xy) \quad = e^x[(1-y^2)\sin(xy) + 2y\cos(xy)]$
$f_{xxy} = e^x[-2y\sin(xy) + x(1-y^2)\cos(xy) + 2\cos(xy) - 2xy\sin(xy)]$